

Generative AI:

10 space

1) Introduction to probabilistic ML.

→ Variational Methods (Latent Variable Models).

→ Mixture Models (GMMs).

→ Variational Auto-Encoder.

→ Diffusion Models (DDPMs).

→ Adversarial family

Generative Adversarial Networks (GANs).

→ Auto-Regressive Models.

Transformer-based LMs.

→ State-Space Models (SSM).

→ Flow-based Models.

① Introduction To probabilistic Machine Learning

→ Model the uncertainty / non-measurable abstracts

Eg :-

a) Classify a body of text as spam or not-spam.

b) Given an image, identify a particular person/object.

Model these system via repeated observations.

→ probabilistic approach

→ Probability :-

A random experiment

- Tossing a coin, rolling a die.
- Taking a picture, writing an email.

Outputs a set of outcomes.

$\Omega = \{H, T, \dots\}$: Coin Toss

$\Omega = \{1, 2, 3, \dots, 6\}$: die roll.

$\Omega = \{\text{pictures of all humans}\}$.

$\Omega = \{\text{texts in the internet}\}$.

Ω : Sample Space: Set of all possible outcomes of a random experiment.

Define probability Measure on Ω :

The idea of a measure.

$$M: \text{Subset} \rightarrow \mathbb{R}^+$$

Consider $\mathbb{R}$, $\mathbb{Z}$, $A, B, C \subset \mathbb{R}^+$

Eg

$$A = [2, 4]$$

$$B = [6, 16]$$

Compare the "size" of A & B , we can define a size measure.

$$A \subset \mathbb{R}, \quad A = [a, b].$$

$$M(A) = |b - a| \quad \text{Leb Measure}$$

Que:- How define a measure that quantifies the idea of "uncertainty" associated with outcomes?

$\mathcal{F}$: Set of subsets of Ω .

$\hookrightarrow$ Event Space.

$$A \in \mathcal{F} \subset \Omega.$$

Define a measure μ .

Define a measure on the elements of $\mathcal{F}$, as follows:

$\mathbb{P} : \mathcal{F} \rightarrow [0, 1] : \text{probability Measure.}$

⇒ properties of $\mathbb{P}$:

a) $\mathbb{P}(A) \geq 0, \forall A \in \mathcal{F}$

b) $\mathbb{P}(\Omega) = 1, \mathbb{P}(\emptyset) = 0.$

c) $A, B \in \mathcal{F}, A \cap B = \emptyset$ Then

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$$

Given there is a random experiment, there is a triplet

$(\Omega, \mathcal{F}, \mathbb{P})$: probability triplet

Sample Space Event Space probability Measure.

In practice, one would not have access to prob. triplet.

Eg: Taking an x-ray image as random experiment.

We get to measure a surrogate of the elements of Ω (Measurements).

We define another function on Ω , To describe the surrogate.

$$X: \Omega \rightarrow \mathbb{R}^d.$$

Random Variable: function from sample space onto real-numbers.

once a function (R.V) is defined on Ω , the probability triplet also gets "transformed".

$$(\Omega, \mathcal{F}, \mathbb{P}) \xrightarrow{X} (\mathbb{R}^d, \mathcal{B}, \mathbb{P}_X).$$

$$X: \Omega \rightarrow \mathbb{R}^d.$$

$\mathcal{F}$ subset of Ω , $\mathcal{B}$ subset of $\mathbb{R}^d$.

$$\mathbb{P}_X(x) = \mathbb{P}(A: x^{-1}[-\infty, x]).$$

$$\begin{aligned} X &\Rightarrow X: \Omega \rightarrow \mathbb{R}. \\ X^{-1}: \mathbb{R} &\rightarrow \Omega \end{aligned}$$

$A \in \mathcal{F}$. This set gets mapped to a subset in $\mathbb{R}$ under X .

Suppose this set $A \in \mathcal{F}$, gets mapped to $[-\infty, x]$ under X .

↳ subset of $\mathbb{R}$.

Then

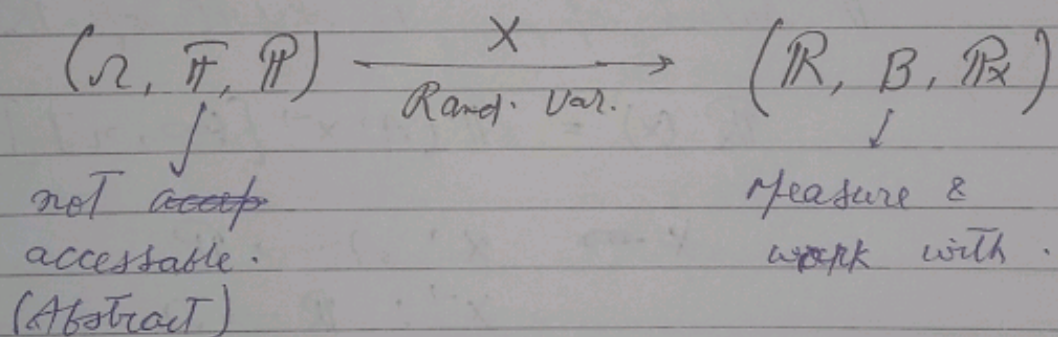
$$P_X(x) = P(A)$$

$$= P\{A \mid x^{-1}((-\infty, x])\}$$

= probability dist. function

prob. distribution function under a random variable X , evaluated at x [$P_X(x)$] is the probability of the events A , that gets mapped to $(-\infty, x]$ under X .

$$P_X(x) = P(A \mid x^{-1}((-\infty, x]))$$



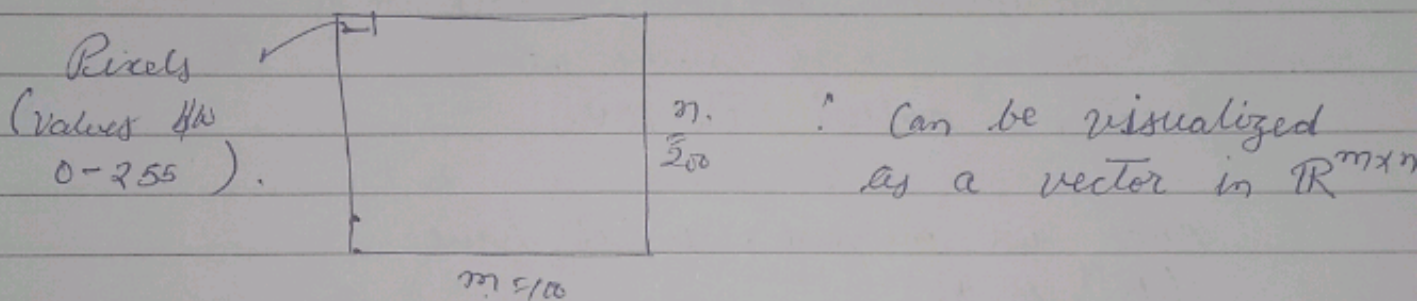
Generative Models

Problem Formation :-

Start with $(\Omega, \mathcal{F}, \mathbb{P})$. define a function
 $X(R.V) : \Omega \rightarrow \mathbb{R}^d$

once a R.V is defined, we'll have
 $(\mathbb{R}^d, \mathcal{B}, \mathbb{P}_X)$: work with this.

a) Consider a problem with images

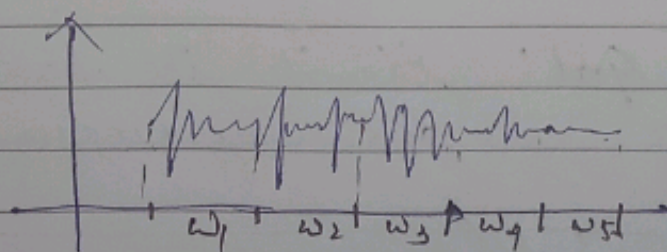


point/vector in $\mathbb{R}^{20000} (d)$

b) Tokens of natural language Text.

Each word is a vector $\mathbb{R}^V$
 $V \rightarrow$ length of a dictionary.

c) Speech Signal :-



Each window is a vector $\mathbb{R}^d$.
 $w \rightarrow$ length of the window.

Summary:- Every datapoint is a vector in an $\mathbb{R}^d$.

$\Rightarrow$ Suppose we have a RV, (x) .

$$X: \Omega \rightarrow \mathbb{R}^d.$$

The datapoints of x (vectors in $\mathbb{R}^d$) are the members of range space of the function called the R.V.

Once we measure, we'll have:
 $(\mathbb{R}^d, \mathcal{B}, \mathbb{P}_x)$.

$\Rightarrow$ The Probability measure ($\mathbb{P}$) completely specifies the underlying sample space.

Goal:- Estimate the underlying probability measure.

The above question translates to the following in practice:

practical Goal: Estimate the probability distribution function

The inputs that are provided:-

Given multiple datapoints:

$$D = \{x_1, x_2, \dots, x_n\} \quad x_i \in \mathbb{R}^d.$$

Eg: N-set of images, paragraphs, code-snippets

x_i is a member of $\mathbb{R}^d$, which is the range space of a $R^V \times X$.

$\Rightarrow$ $\mathcal{F}$ a sample space Ω &
the cross corresponding prob measure $\mathbb{P}$.

& a distribution function $\mathbb{P}_X$.

$$D = \{x_i\}_{i=1}^N \stackrel{\text{i.i.d}}{\sim} \mathbb{P}_X,$$

$$x_i \in \mathbb{R}^d.$$

Given D , estimate $\mathbb{P}_X$.

$\Rightarrow$ In generative models, the problem is to not only estimate $\mathbb{P}_X$, but also to learn to sample from it.

$\Rightarrow$ Sampling: process of conducting the experiment that gave rise to Ω .

Q → Problem Formulation for GenAI :-

Given a dataset D .

$$D = \{x_1, x_2, \dots, x_N\} \stackrel{\text{i.i.d}}{\sim} P_X \text{ (unknown)}$$

$$x_i \in \mathbb{R}^d.$$

Objective: Estimate P_X & sample from it.

Q → Framework for solving the GenAI problem :-

- i) Assume a parametric form on the unknown distribution function.

P_θ where θ are the set of parameters.
↳ "Model". Eg: Neural Networks

- ii) Define / Measure a divergence metric b/w P_X & P_θ .

$D(P_X \| P_\theta)$: Measures how far are P_X & P_θ

- iii) Adjust the parameters θ , st.

$D(P_X \| P_\theta)$ is minimized

$$\theta^* = \underset{\theta}{\operatorname{argmin}} [D(P_X \| P_\theta)]$$

iv) once Π_x is estimated, use it to sample.